

Supplementary Methods

Extended methodological detail for *Retained Spectral Energy Is Not Reconstruction Fidelity: A Verified Re-Analysis of DFT Coefficient Selection*. Section numbers refer to the main manuscript.

S1. Data recovery in full

The source data of the 2013/2014 study are unavailable. That work prints the real and imaginary parts of its transform coefficients for $k = 0, \dots, 14$ of a real sequence with $N = 26$. The degree-of-freedom count is exact: bin 0 is real (1), bins 1–12 are complex (24), and bin 13 is real because N is even and $13 = N/2$ is the Nyquist bin (1), totalling 26. Hermitian symmetry $X[N-k] = X^*[k]$ supplies bins 14–25.

The coefficients are expressed in the earlier work's own convention, which uses a *positive* analysis exponent with $1/\sqrt{N}$ scaling. The recovery therefore applies the synthesis matrix of *that* convention, not of the convention adopted for the present study. Applying the wrong one yields a time-reversed sequence, which is a useful self-check: it does not match the printed samples.

Validation evidence

Three independent conditions were checked before the recovered data were used for anything.

(i) Agreement with printed samples. The 15 sample values printed in the earlier work are reproduced exactly after rounding. Before rounding, the residual has mean -0.0627394 against -0.0627497 predicted from the DC coefficient being printed to four significant figures. The residual is *not* exactly constant: its spread is 1.17×10^{-3} , attributable to the remaining coefficients being printed to three decimals. This distinction is recorded because an earlier informal analysis described the residual as a constant offset, and the more careful statement is that its *mean* is explained by DC rounding to within 10^{-5} .

(ii) Internal consistency of the printed coefficients. The printed coefficient at $k = 14$ equals the conjugate of that at $k = 12$, and the printed imaginary part at the Nyquist bin is 2.154×10^{-12} . Both are conditions that a real length-26 sequence must satisfy, and neither was used in constructing the recovery, so they are genuine checks.

(iii) Integrality. Recovered values lie within 0.064 of integers throughout, consistent with integer-valued source data and with the magnitude of the DC-rounding offset.

| **What the recovery is not.** It is a reconstruction of the source data, not the original file. The
| physical identity, measurand, units and sampling rate of the signal are not stated in the earlier
| work and are not recoverable from it. No provenance is inferred or invented, and all analysis treats
| the sequence as dimensionless.

S2. Transform conventions

One convention is used throughout the manuscript, its figures, tables, code and this supplement: the unitary pair with a negative analysis exponent and $1/\sqrt{N}$ scaling on both directions. The legacy transform of the earlier work is retained in the code solely so that its published coefficient tables can be reproduced exactly; it is never used for new analysis. It satisfies $y_e[k] = X[(N-k) \bmod N]$, i.e. it is the frequency-reversed conjugate of the standard DFT, which is asserted as a validation gate.

S3. Conjugate-group retention

All retention sets are built from whole conjugate groups. For $N = 26$ the groups are $\{0\}$, $\{13\}$ (both self-conjugate) and the twelve pairs $\{k, 26-k\}$ for $k = 1, \dots, 12$. Retaining whole groups guarantees a real-valued reconstruction. The reconstruction function asserts this and raises if the imaginary residual is not negligible, rather than silently taking a real part — a non-conjugate mask is a specification error, not a rounding issue.

S4. Noise experiment design

SNR is defined on the mean-removed power of the baseline. Scaling noise to the DC-inflated total power would make the nominal SNR meaningless for exactly the reason this study documents: at $r = 3.70$ the total power is 14.7 times the fluctuation power, so a nominal "0 dB" defined on total power would in fact be about +11.7 dB on the quantity of interest.

Errors are measured against the *clean* baseline rather than the noisy observation, because the task is recovery of the underlying signal. Each SNR level draws from a distinct reproducible child stream derived from the master seed, so levels are independent yet exactly reproducible.

S5. Statistical policy

Deterministic experiments are reported as exact values with no dispersion, because they have none: repeating them yields bit-identical results.

The noise experiment uses 100 realizations per level and is reported as mean, standard deviation and a 95% confidence interval for the mean under the normal approximation. No p-value is reported and no significance test is performed. This is a deliberate choice, not an omission: no hypothesis in this study takes the form of a comparison against a null that a test would illuminate. The quantity of interest is an effect size — the gap between retained energy and reconstruction error — and it is reported directly, in the units in which it matters. Normality tests were likewise not performed, as they would serve the appearance of rigor rather than any inference actually drawn.

| The reported intervals are confidence intervals for the *mean across realizations*. They are not
| prediction intervals for a single realization and must not be read as such.

S6. Window characterization

Main-lobe widths and sidelobe levels are measured from a densely zero-padded frequency response (oversampling factor 512), not from the length-26 transform. With $N = 26$ the transform has only 14 unique bins, which cannot resolve a main lobe a few bins wide, let alone sidelobe structure. An earlier implementation that measured lobes on the native transform reported spurious 0 dB sidelobes, because for a real signal it found the conjugate mirror of the main lobe; the defect was caught by comparison against published values and is recorded here because the failure mode is easy to reproduce.

Measured peak sidelobe levels agree with the published catalog to within 1.5 dB for all four windows, providing an external check on the implementation that does not depend on any code in this package.

S7. Provenance of each element

Referenced from Appendix C of the manuscript. Material inherited from the 2013/2014 work, results recomputed and confirmed, corrections made, and results new to this study.

Category	Content
Inherited from [11]	The 26-sample sequence; the real-form transform pair; the low-pass retention scheme; the originating question about coefficients per energy percentage
Recomputed and confirmed	E, H, E1, $100 \cdot E1/E$, E2, $100 \cdot E2/E$, $0.95E$, and the coefficient tables for $k = 0-14$. All nine reproduce; see manuscript Table 2
Corrected	Analysis-kernel sign and spectrum orientation; normalization convention; “inverse of the function” $\rightarrow$ modulation / $N/2$ spectral shift; $(x6)^2 \rightarrow x6 ^2$; the interpretation of the retention figure
New to this study	Data recovery with existence and uniqueness; equations (4) and (7)–(10); Experiment A; the AC-basis analysis; the selection comparison with exhaustive optimality check; the DCT comparison; leakage and window experiments; the noise study; the precision sweep; the reporting protocol
Not claimed	Any new transform theory; identification of the DC-dominance phenomenon, which is established in the biomedical compression literature; any prevalence estimate; any claim about a specific application domain

S7. Validation gates

The pipeline runs 32 numerical gates and aborts if any fails. Tolerances are fixed in `validation.py` and are never relaxed to make a check pass. The gates cover transform correctness, direct-DFT/FFT agreement, Hermitian and Nyquist structure, the convention discrepancy, Parseval

at double precision, the DC-share identity, DCT orthonormality, reconstruction realness, all nine historical quantities, the three exact relations, window sidelobes against the literature, and selection-mask sanity.